

PAPER_202: UQFF Reionization, BBN, Recombination, and Cosmic Dawn Physics

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{bi}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$$

Abstract

This paper presents the UQFF buoyancy formalism applied to cosmic dawn and reionization physics: baryon-to-photon ratio (?), deuterium bottleneck during Big Bang Nucleosynthesis (BBN), CMB angular power spectrum, recombination optical depth, ionization fraction evolution, HII bubble growth rate, and Jeans mass/wavelength for fragmentation. These collectively span the redshift range $z \sim 1100$ (recombination) through $z \sim 5$ (end of reionization). Each F_UBii and Um variant is rigorously derived from observational anchor equations.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Baryon-to-Photon Ratio (?)

$\eta = n_b/n_\gamma = 6.08 \times 10^{-10}$ (Planck 2018 + BBN fit)

$F_{UBii, \eta} = F_{rel} \times (\eta = n_b/n_\gamma / E_{LEP}) \times Q_{wave} \times [\text{Neff modification}]$

$U_{m, \eta}(t) = \mu(\eta_{vac}) \cdot (1 - e^{-\eta t}) \cdot \eta$

Physical origin:

$n_\gamma = 410 \text{ cm}^{-3}$ (CMB photon number density today)

Fit anchor: Primordial D, ^3He , ^4He , ^7Li abundances

η determines all light element yields (standard BBN)

UQFF role: η sets vacuum energy scale through η_b/η_γ at nucleosynthesis

2. Deuterium Bottleneck (BBN Timing)

Deuterium bottleneck timescale:

$t_D \sim \nu(3/(32\pi G \eta_{rad})) \sim 180 \text{ s}$ at $T \sim 0.1 \text{ MeV}$

$$\tau_{\text{rad}} = p^2 k T^4 / (30 h^3 c^5) \cdot g_* \quad (\text{effective DOF } g_* \sim 10 \text{ at D-formation onset})$$

Sequence:

$T \sim 1 \text{ MeV}$: neutron freeze-out ($n/p \sim 1/7$)

$T \sim 0.1 \text{ MeV}$ ($t \sim 180 \text{ s}$): D photodissociation ends ? 4He chains proceed

Yield: $Y_P \sim 0.247$ (mass fraction 4He)

$$F_{\text{UBii,deb}} = -F_{\text{rel}} \times (t_D / E_{\text{LEP}}) \times Q_{\text{wave}} \times [g_*(T) \text{ variation}]$$

$$U_{\text{m,deb}}(T) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau_t}) \cdot [\text{Weak freeze at } T \sim 1 \text{ MeV; D photodissociation below}]$$

UQFF calibration: ? calibrated to UQFF $k_?$ parameter through BBN freeze-out

$k_? \sim 10^{113}$ (small-scale vacuum fluctuation coupling)

$\tau k_?(\text{CIA cross-section refit}) \sim 7.25 \times 10^8$ (fractional shift from H2 CIA data)

3. CMB Angular Power Spectrum

$$C_l = (2/p) \cdot \tau k^2 dk \cdot P(k) \cdot |\tau_l^T(k)|^2$$

$P(k) \propto k^{n_s-4}$ (primordial, $n_s \sim 0.965$ Planck 2018)

$\tau_l^T(k)$ = transfer function:

- Large scales ($l < 100$): Sachs-Wolfe plateau $C_l \sim \text{constant}$
- Acoustic peaks ($l = 220, 540, 800 \dots$): baryon-photon oscillations
- Damping tail ($l > 1000$): Silk damping

$$F_{\text{UBii,cmb}} = F_{\text{rel}} \times (C_l \cdot l(l+1)/(2p) / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$$U_{\text{m,cmb}}(k) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau_t}) \cdot \text{Transfer } \tau_l^T: \text{Sachs-Wolfe + acoustic oscillations}$$

Calibration anchor:

$D_l = l(l+1)C_l/(2p) \times T_0^2$? peak at $l \sim 220$ matches $0_{\text{tot}} = 1$

UQFF role: vacuum energy $\tau c^2/3$ term in g_{UQFF} sets acoustic horizon

4. Recombination Optical Depth

$$t(z) = \tau_z^8 n_e(z') \cdot s_T \cdot c \cdot |dt/dz'| dz'$$

$s_T = 6.652 \times 10^{-29} \text{ m}^2$ (Thomson cross-section)

$$n_e(z) = x_e(z) \cdot n_H(z) = x_e(z) \cdot n_{b,0} \cdot (1+z)^3$$

Recombination: $z_{\text{rec}} \sim 1100$? $t(z_{\text{rec}}) \sim 1$ (photon decoupling)

Reionization: $z_{\text{re}} \sim 7.7$? optical depth $t_{\text{re}} \sim 0.054$ (Planck 2018)

$$F_{\text{UBii,recomb}} = -F_{\text{rel}} \times (t(z) / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$$U_{\text{m,recomb}}(z) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau_t}) \cdot \tau n_e \cdot s_T \cdot c \cdot dt$$

Physical context:

After z_{rec} , photons decouple and stream freely (CMB)

Surface of last scattering has $\tau/T \sim 10^{-5}$ (measured by Planck)

5. Reionization: Ionization Fraction Evolution

$$dx_e/dt = \Gamma_{\gamma} \cdot e_{\text{esc}} \cdot f_{\star} - a_B \cdot n_e^2 \cdot C$$

where:

$$\begin{aligned}\Gamma_{\gamma} &= (1+z)^3 \cdot n_b \cdot \Gamma_{\gamma, \text{eff}} && \text{(ionizing photon rate)} \\ e_{\text{esc}} &\sim 0.1-0.3 && \text{(photon escape fraction from galaxies)} \\ f_{\star} &\sim 0.05-0.2 && \text{(star formation efficiency in halos)} \\ a_B &= 2.6 \times 10^{-13} \text{ cm}^3/\text{s} && \text{(recombination coefficient, case B, } T \sim 10^4 \text{ K)} \\ C &= \bar{n}_H^2 / n_H^2 && \text{(clumping factor, } C \sim 3)\end{aligned}$$

$$F_{\text{UBii, ion}} = F_{\text{rel}} \times (\Gamma_{\gamma} \cdot e_{\text{esc}} \cdot f_{\star} / E_{\text{LEP}}) \times Q_{\text{wave}} \times [\text{recombination subtracted}]$$

$$U_{\text{m, ion}}(t) = \mu(\bar{n}_{\text{vac}}) \cdot (1 - e^{-\Gamma_{\gamma} t}) \cdot a_B \cdot n_e^2 \cdot C \, dt$$

Reionization history:

- $z \sim 20-30$: first stars ionize H around them (PopIII)
- $z \sim 7-9$: bulk reionization, x_e rises from ~ 0.1 to ~ 1
- $z \sim 5-6$: completion (Gunn-Peterson trough in quasar spectra)

6. HII Bubble Growth During Reionization

$$dN_b/dt = \Gamma_{\gamma, \text{eff}} \cdot e_{\text{esc}} - a_B \cdot n_H \cdot (4/3 \pi R_b^3)$$

Simplified overlap model (Stromgren sphere analog):

$$R_b(t) = (3 \Gamma_{\gamma} \cdot t / (4 \pi \cdot n_H))^{1/3} \quad \text{(linear ionization front growth)}$$

Γ_{γ} = rate of ionizing photons emitted (from galaxy SFR)

n_H = neutral H density (z-dependent)

$$F_{\text{UBii, bub}} = F_{\text{rel}} \times (R_b(t) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (\Gamma_{\gamma} \cdot x_e)$$

$$U_{\text{m, bub}}(t) = \mu(\bar{n}_{\text{vac}}) \cdot (1 - e^{-\Gamma_{\gamma} t}) \cdot [\text{Expand for moving ionization front}]$$

Physical context:

Bubble merger & percolation at $x_{\text{HI}} < 0.1$ & full reionization

UQFF buoyancy at bubble edge acts as expansion driver ($F_{\text{UBii, bub}} > 0$)

7. Jeans Length and Mass (Fragmentation)

Jeans length:

$$\lambda_J = p^{1/2} \cdot c_s / (G \cdot \rho)^{1/2}$$

Jeans mass:

$$M_J = (p/6) \cdot \lambda_J^3 = (5k_{\text{BT}} / (G \mu m_H))^{3/2} \cdot (3/(4 \pi \rho))^{1/2}$$

Stability condition: $\lambda > \lambda_J$ & gravitational collapse

$$\text{Dispersion relation: } \omega^2 = c_s^2 \cdot k^2 - 4 \pi G \rho$$

$$\text{Unstable: } \gamma^2 < 0 \quad \gamma k < k_J = 2p/\gamma_J$$

$$F_{\text{UBii,jeans}} = -F_{\text{rel}} \times (\gamma_J / E_{\text{LEP}}) \times Q_{\text{wave}} \times (\text{collapse onset time } t \sim 1/v(G))$$

$$U_{\text{m,jeans}}(T) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot [\text{Perturb: } \gamma^2 = c_s^2 \cdot k^2 - 4pG]$$

Physical context:

First (PopIII) stars: $T \sim 200 \text{ K}$, cloud $M \sim 100\text{--}1000 M_\odot$

Present-day GMCs: $T \sim 10\text{--}30 \text{ K}$ $M_J \sim 1\text{--}10 M_\odot$

UQFF buoyancy at Jeans scale: $F_{\text{UBii,jeans}}$ acts as tidal disruption force

8. Alfvén Wave and Turbulent Cascade

Alfvén wave velocity:

$$v_A = B/v(4p) \quad (\text{for } B \text{ in Gauss, } \gamma \text{ in g/cm}^3)$$

Anisotropic cascade (Goldreich-Sridhar):

$$k_\perp/k_\parallel \sim (k_\perp \cdot l_A)^{1/3} \quad (\text{scale-dependent anisotropy})$$

$$F_{\text{UBii,alf}} = F_{\text{rel}} \times (v_A / E_{\text{LEP}}) \times Q_{\text{wave}} \times (B \cdot \gamma v_A) \times e^{-t/t_{\text{eddy}}}$$

Turbulent energy cascade (Kolmogorov in ISM):

$$e = v_l^3/l = \text{constant} \quad (\text{energy transfer rate per unit mass})$$

$$v_l \propto l^{1/3} \quad (\text{velocity-scale relation})$$

$$\text{Power spectrum: } E(k) \propto k^{-5/3}$$

$$F_{\text{UBii,turb}} = F_{\text{rel}} \times (e^{2/3} \cdot l^{-2/3} / E_{\text{LEP}}) \times Q_{\text{wave}} \times e^{-kl/k_J}$$

$$U_{\text{m,alf}}(B) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot v_A(B)$$

$$U_{\text{m,turb}}(l) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot (\gamma \cdot v_l^3/l)$$

9. Ionization Parameter and Feedback Coupling

Ionization parameter:

$$U = Q_H/(4p \cdot r^2 \cdot n_H \cdot c)$$

where:

Q_H = number of ionizing photons per second (from AGN or OB stars)

r = distance from ionizing source

$\log U \sim -2 \text{ to } -3$ for NLR, $-1 \text{ to } 0$ for BLR

AGN feedback coupling efficiency:

$$e_f = E_{\text{kin}}/(\gamma_{\text{acc}} \cdot c^2) \sim 0.05\text{--}0.1$$

$$E_{\text{kin}} = (1/2) \gamma_{\text{out}} \cdot v_{\text{out}}^2$$

Only $e_f \times 10^{-5}$ per $M_\odot$ needed to match $M_{\text{BH}}\text{--}s$

$$F_{\text{UBii,upar}} = F_{\text{rel}} \times (U \cdot n_H \cdot c / E_{\text{LEP}}) \times Q_{\text{wave}} \times (Q_H/r^2)$$

$$F_{\text{UBii,coup}} = F_{\text{rel}} \times (e_f \cdot \gamma_{\text{acc}} \cdot c^2 / E_{\text{LEP}}) \times Q_{\text{wave}} \times [0.05\text{--}0.1]$$

$$U_{\text{m,upar}}(r) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot (Q_H/(4\pi r^2 n_H c))$$

$U_{m,coup}(?) = \mu(?_{vac}) \cdot (1 - e^{-\tau}) \cdot e_f$ (couple fraction to kinetic energy)

10. Cosmological Timeline Summary

Epoch	Redshift	UQFF Process	F_UBii Variant
Inflation	$z \sim 10^{27}$	Inflaton field $V(?)$	F_UBii,inf
Planck epoch	$z \sim 10^{32}$	LQC bounce	F_UBii,bounc
Neutron freeze-out	$z \sim 10^{10}$ (T~1 MeV)	Weak interactions	F_UBii,eta
BBN	$z \sim 10^9$ (t~180 s)	D bottleneck	F_UBii,deb
Recombination	$z \sim 1100$	Photon decoupling	F_UBii,recomb
Cosmic dawn	$z \sim 20-30$	First stars, first HII	F_UBii,ion
Reionization	$z \sim 6-9$	HII bubble percolation	F_UBii,bub
Present	$z \sim 0$	Structure + feedback	All F_UBii

11. References

- grok_share_7514fe.txt lines 6070-6090 (BB_C_Equations items 1310-1350)
- grok_share_7514fe.txt lines 6300-6500 (full reionization/BBN/CMB catalogue)
- PAPER_199: F_UBii Taxonomy Part 2 (Cosmological)
- PAPER_200: Um Universal Magnetism Catalogue
- Planck 2018 Collaboration: CMB and reionization parameters

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see uqff_lagrangian_derivation.py).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2) \phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.123 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 83, \quad n_{\text{channel}} = 21/26$$

Since $p_{\text{DVP}} = 83$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10⁴ yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.123	✓ Threshold-consistent
DVP prime BSH layers	$p_k \in \{2,3,\dots,113\}$ 26 harmonic terms	$p_{\text{DVP}} = 83$ $j = 1\dots 26,$ $\cos(2\pi j/26)$	✓ Resonant ✓ Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}} \text{ suppression}$	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.